UNIT-3
BIOCHEMISTRY & MOLECULAR BIOLOGY

Carbohydrates

Protein structure

Structure of DNA and its differences with RNA

Replication

Transcription

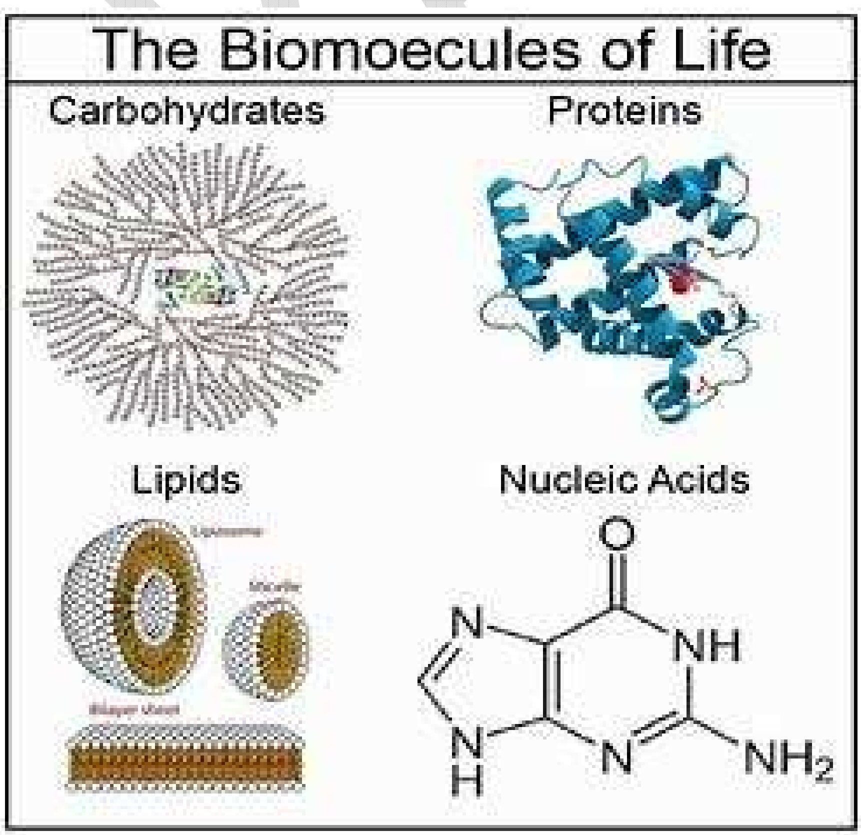
Translation

Biomolecules – An Introduction

Organic compounds that make up the very backbone of a cell in living organisms are known as biomolecules. These could include a variety of compounds in varying proportions such as carbohydrates, fats and proteins. All of these biomolecules have vital functions to perform, and they are manufactured within the body.

While organic compounds may consist of straight-chain carbons, others can be cyclic rings, branched chains or combinations of any of these types. They have varying features, chemical properties and structures; this, in turn, leads to the difference in physical properties like boiling and melting points, solubility in water. Depending on their affinity towards water, biomolecules can be categorised further into hydrophilic and hydrophobic.

Biomolecules are categorized into four types. They are:



Carbohydrates

They are the aldehyde or ketone derivatives of polyhydric alcohols or as the compounds that yield these derivatives on hydrolysis.

Proteins

Organic compounds present in our diets in generous amounts and made of amino acids are called proteins. They consist of long-chain monomers bonded together with the help of polypeptide bonds. Therefore, proteins are also sometimes called polypeptides.

Lipids

These are a group of hydrophobic compounds that include fats, oils, steroids, phospholipids and glycerol.

Nucleic Acids

The smallest fundamental units of our genetic information, also called genes, consist of building blocks called DNA and RNA. These constitute the genetic material of our bodies and are a combination of nitrogenous bases, molecules of sugar and a phosphate group.

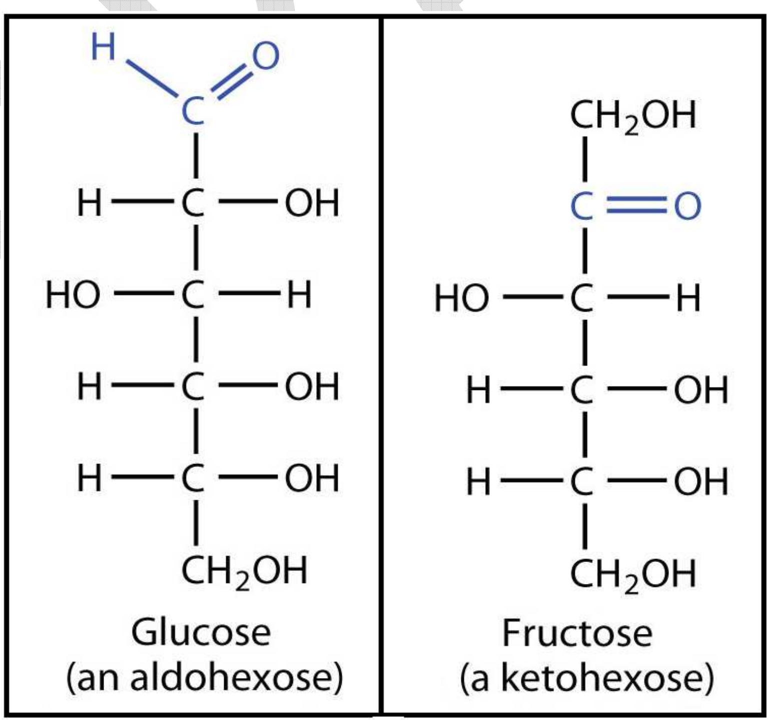
CARBOHYDRATES

Carbohydrates may be defined as polyhydroxy aldehydes or ketones or as substances that yield one of these compounds on hydrolysis.

These are the organic compounds consisting of carbon, hydrogen and oxygen, the last two in a 2:1 ratio as in water. Carbohydrates are a group of universally occurring compounds having the general formula $(CH_2O)_{n,}$. Carbohydrates serve as the chief source of energy in the food of many animals. The carbohydrates are widely distributed both in animal and plant tissues. In animal cells they occur chiefly in the form of glucose and glycogen, where as in plants cellulose and starch are the main representatives. Carbohydrates are also important structural components.

STRUCTURE

Carbohydrates are aldehydes and ketone derivatives of Polyhydroxy alcohols. Each carbohydrate therefore contains an aldehyde or a ketonic group and is known as aldose or a ketose.



CLASSIFICATION

On the basis of the number of monomeric units present, carbohydrates are classified as follows:

- Monosaccharides
- Oligosaccharides
- Polysaccharides

MONOSACCHARIDES (Mono = Single; Saccharide = Sugar)

Monosaccharides are the simplest carbohydrates which cannot be hydrolyzed to smaller molecules. The monosaccharides are divided into two categories:

1) Based on the functional group
2) Number of carbon atoms.

When the functional group is an aldehyde, they are known as aldoses Ex: glyceraldehyde, glucose.

When the functional group is a keto (-c=o), they are referred as ketoses ex: fructose.

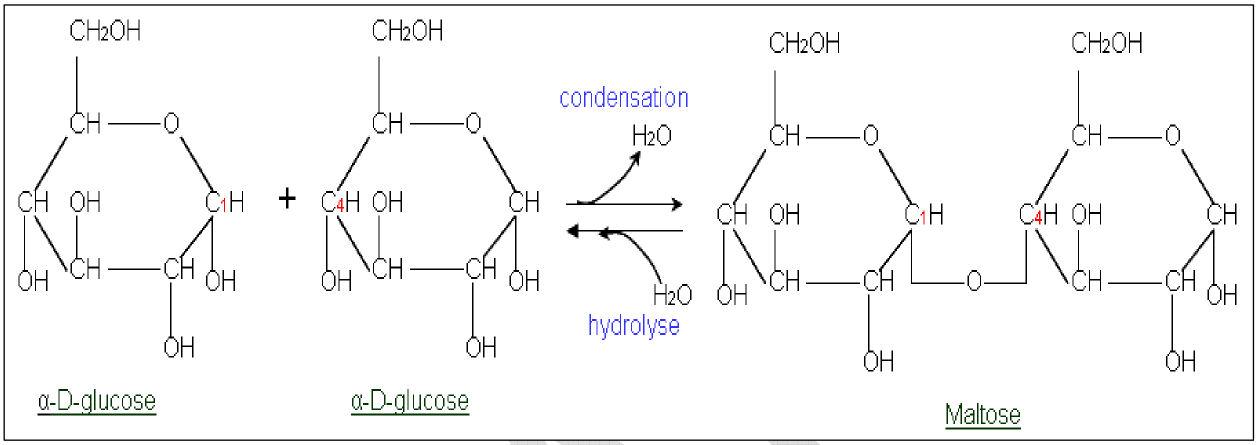
Based on the number of carbon atoms, the monosaccharides are regarded as trioses (3c), tetroses (4c), pentoses (5c), hexoses (6c), heptoses (7c) and octoses (8c).

Classification of Monosaccharides

No. of Carbon	Type of sugar	Aldoses	Ketoses
3	TRIOSES	Glyceraldehydes	Dihydroxyacetone
4	TETROSES	Erythrose	Erythrulose
5	PENTOSES	Ribose, Xylose	Ribulose, xylulose
6	HEXOSES	Glucose, Galactose	Fructose
7	HEPTOSES	Glucoheptose	Sedoheptulose

OLIGOSACCHARIDES

The Oligosaccharides are those Carbohydrates which on hydrolysis give two to ten simple monosaccharide molecules. During union of monosaccharide units water molecule is eliminated and the units are linked through an oxygen bridge known as O-**glycosidic linkage**.



They are a large category and further divided into various subcategories.

Disaccharides: These give two units of the same or different monosaccharides on hydrolysis. Ex are sucrose, lactose and maltose.

	Sucrose	**Lactose**	**Maltose**
Commercial name	Table sugar Cane sugar Invert sugar	Milk sugar	Malt sugar
Composition	Glucose + Fructose	Glucose + Galactose	Glucose + glucose
Hydrolysing enzyme	Sucrase/ Invertase	Lactase	Maltase
Sources	sugarcane	Human milk, cow milk	Barley

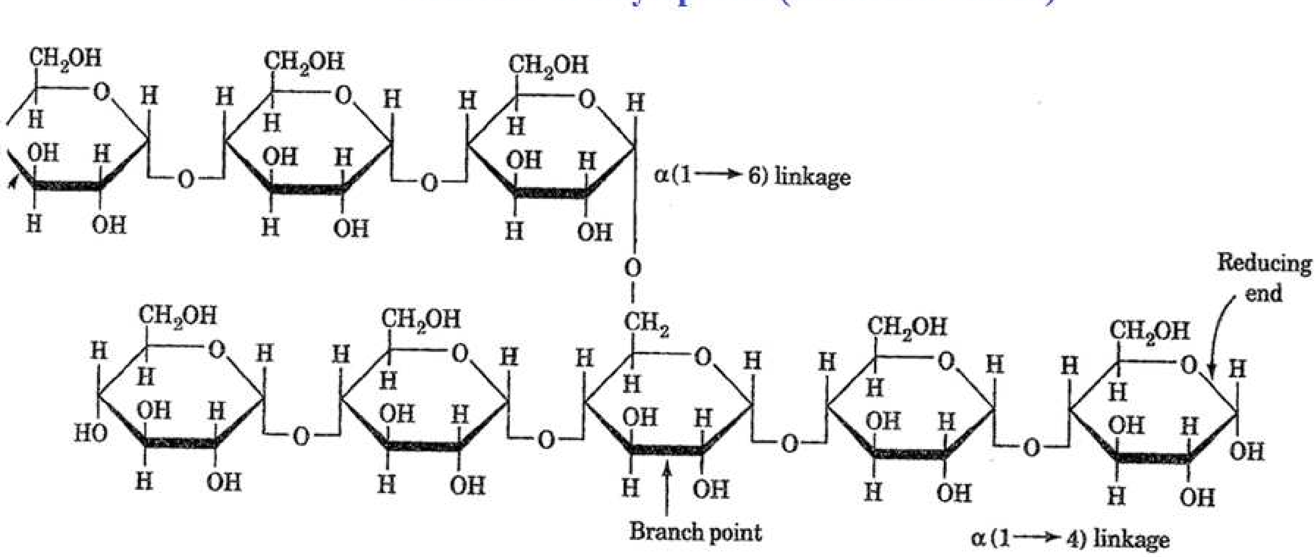
Trisaccharides: Carbohydrates that on hydrolysis gives three molecules of monosaccharides, whether same or different. An example is Raffinose.

Tetrasaccharides: And as the name suggests this carbohydrate on hydrolysis give four molecules of monosaccharides. Stachyose is an example.

POLYSACCHARIDES (GLYCANS)

Majority of the carbohydrates found in nature occur as polysaccharides of high molecular weight. Polysaccharides yield more than 10 molecules of monosaccharides on hydrolysis. Some Polysaccharides are linear polymers and others are highly branched. In all cases the linkage that unites the monosaccharide units is the glycosidic bond. They are tasteless, colourless and insoluble in water. Because of insolubility and large size they form colloidal solutions and will not pass across natural animal membranes. They are represented by a general formula $(C_6H_{10}O_5)_n$.

Structure of amylopectin (Branched Starch)



Polysaccharides are of two types

a. **Homopolysaccharides (Homoglycans):** They contain monosaccharide units of a single type. Ex: starch, inulin, cellulose, glycogen

b. **Heteropolysaccharides (Hetroglycans):** They possess two or more different types of monosaccharide units or their derivatives. Ex: chitin, heparin, chondroitin sulfate

FUNCTIONS OF CARBOHYDRATES

Carbohydrates participate in a wide range of functions.

1) They are the most abundant dietary source of energy (4 cal/g) for all organisms. About 60% of the total energy requirement of man is provided by the breakdown of carbohydrates.
2) Carbohydrates play a key role in the metabolism of amino acids and fatty acids.
3) Carbohydrates serve as an important structural material in some animals and in all plants, where they constitute the cellulose frame work, monosaccharides are important constituents of nucleic acids, co- enzymes, flavor-proteins etc. Heparin prevents the clotting of blood. Chondroitin sulphates are found in cornea, cartilage tendons, skin, heart valves and saliva.
4) Carbohydrates are utilized as raw materials for several industries.
   Ex: paper, plastic, textiles, alcohol etc.

The physiological role of some carbohydrates is given below

MONOSACCHARIDES	
Ribose	It is a structural element of ATP, nucleic acids(RNA) and coenzymes (NAD, NADP, Flavo proteins)
Deoxyribose	It is a structural element of Nucleosides, Nucleotides and DNA
Glucose	It is the sugar of the body serving the tissue as a major metabolic fuel.
Galactose	It is synthesized in mammary glands and hydrolyzed to make the lactose of milk.
Fructose	It occurs naturally in fruit juices and honey.
DISSACHARIDES	
Sucrose	Most commonly used table sugar.
Lactose	Major sugar in milk.
Maltose	Product of starch hydrolysis.
POLYSACCHARIDES	
Starch	Reserve food in plants. Ex. cereals, potatoes, legumes, roots, tubers etc
Glycogen	Reserve food in animals (hence called as animal starch). It is present in

	high concentration in liver, followed by muscle, brain etc.
Cellulose	It is the chief constituent of plant cell wall.
Chitin	Structural Polysaccharide in the exoskeleton of insects and crustaceans.
Inulin	It is a polymer of fructose found in *Dahlia* bulbs, garlic, onion etc. Used for assessing kidney function through measurement of glomerular filteration rate (GFR).

Objective type Questions:

1) The sugar present in the blood
   a) **Glucose**
   b) Fructose
   c) Galactose
   d) Manose

2) Carbohydrates which are not found in plants
   a) Inulin
   b) Cellulose
   c) Starch
   d) **Galactose**

3) Major metabolic fuel of the tissue is
   a) Galactose
   b) Sucrose
   c) Fructose
   d) **Glucose**

4) Monosaccharide units are linked with each other with the help of
   a) Hydrogen bonds

b) Covalent linkages
c) Ionic bonds
d) Glycosidic linkages

5) Maltose is a
a) **Disaccharide**
b) Monosaccharide
c) Oligosaccharide
d) Polysaccharide

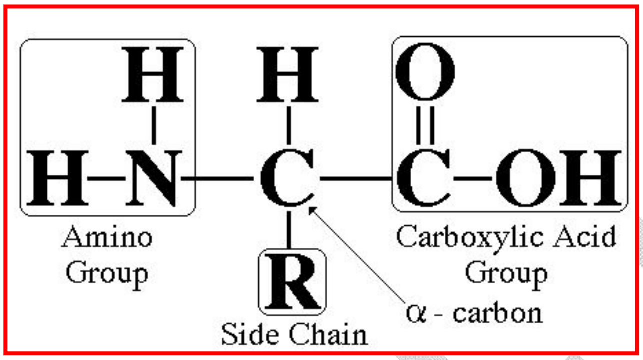
6) One of the following statement is not correct with regards to polysaccharides
a) They are represented by a general formula $(C_6H_{10}O_5)_n$
b) They are tasteless, colourless and insoluble in water
c) They form colloidal solutions
d) On hydrolysis they give 2 to 10 monosaccharide units

7) Animal starch is
a) Glucose
b) **Glycogen**
c) Chitin
d) None

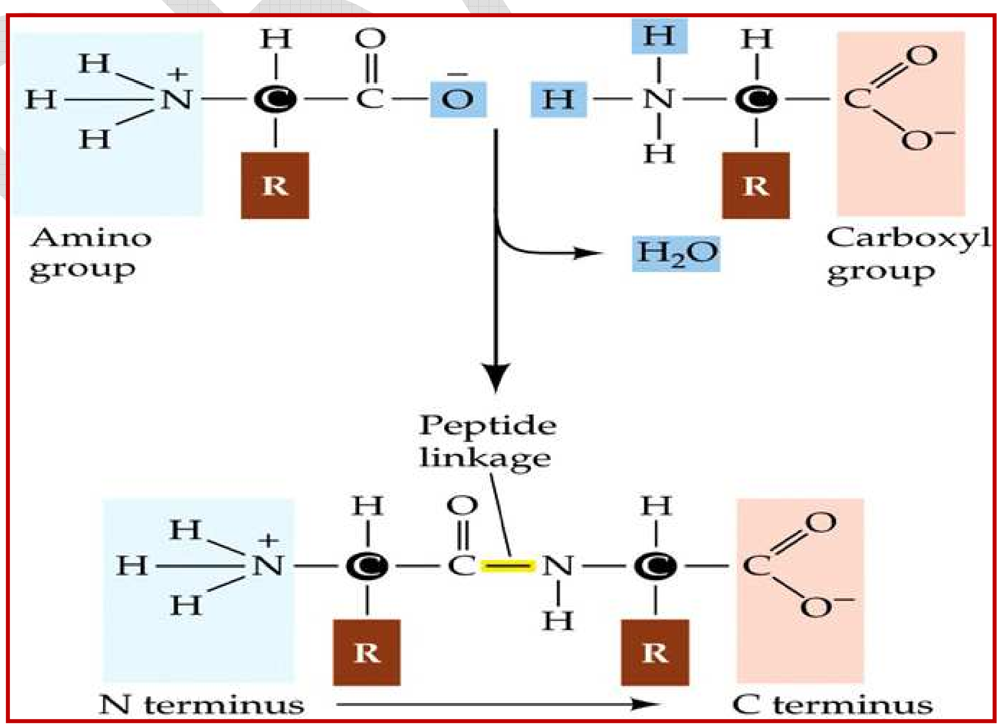
8) Important structural polysaccharide present in the exoskeleton of some invertabrates is
a) Inulin
b) Glycogen
c) **Chitin**
d) Cellulose

PROTEINS

The building blocks of proteins are amino acids, which are small organic molecules that consist of an alpha (central) carbon atom linked to an amino group, a carboxyl group, a hydrogen atom, and a variable component called a side chain.



Proteins are built from a set of only **twenty amino acids**, each of which has a unique side chain. Within a protein, multiple amino acids are linked together by **peptide bonds**, thereby forming a long chain. Peptide bonds are formed by a biochemical reaction that extracts a water molecule as it joins the amino group of one amino acid to the carboxyl group of a neighboring amino acid.

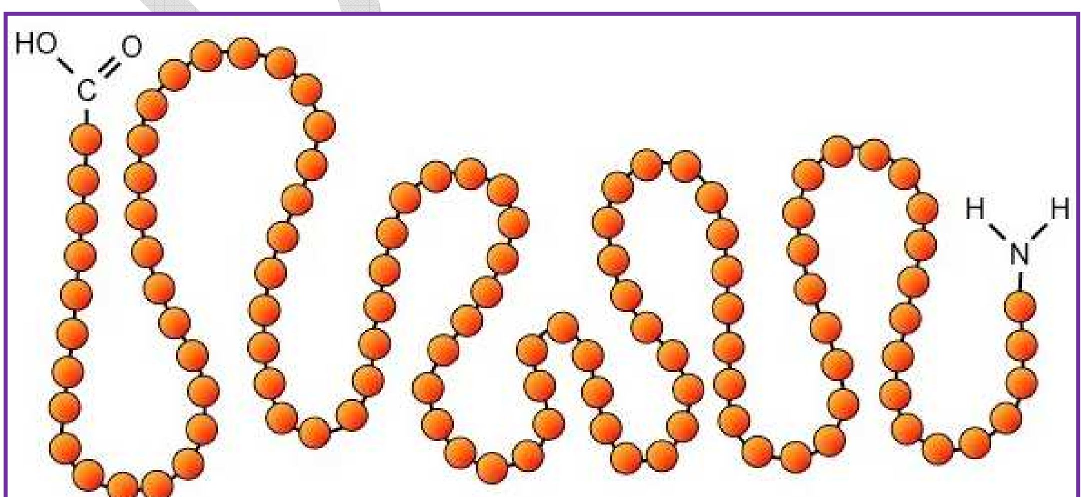


STRUCTURE OF PROTEIN MOLECULES

Proteins are natural polymers that are ranked first amongst the chemical substances essential for the growth and maintenance of life. These are the compounds of carbon, oxygen, hydrogen and nitrogen. Some proteins contain additional elements such as phosphorus, iron, zinc and copper. The proteins are very large compact molecules often referred to as macro molecules. Each protein is made up of numerous simpler units called the amino acids which are joined together by peptide bonds. When many amino acids are linked together, the combination is called polypeptide. The polypeptide chains unite to form very large molecule called proteins. Four basic structural levels are assigned to proteins. These are primary, secondary, tertiary and quaternary structures.

1) Primary structure

The primary structure of a protein refers to the linear sequence of amino acids in the polypeptide chains. Every protein is determined by the sequencing of the amino acids. The formation and ordering of these amino acids in proteins are extremely specific. If we alter even one amino acid in the chain it results in a non-functioning protein or what we call a gene mutation. In the year 1953, Frederick Sanger worked out the sequence of amino acids in the polypeptide chain of the hormone insulin. Thus, insulin became the first protein molecule to be sequenced.



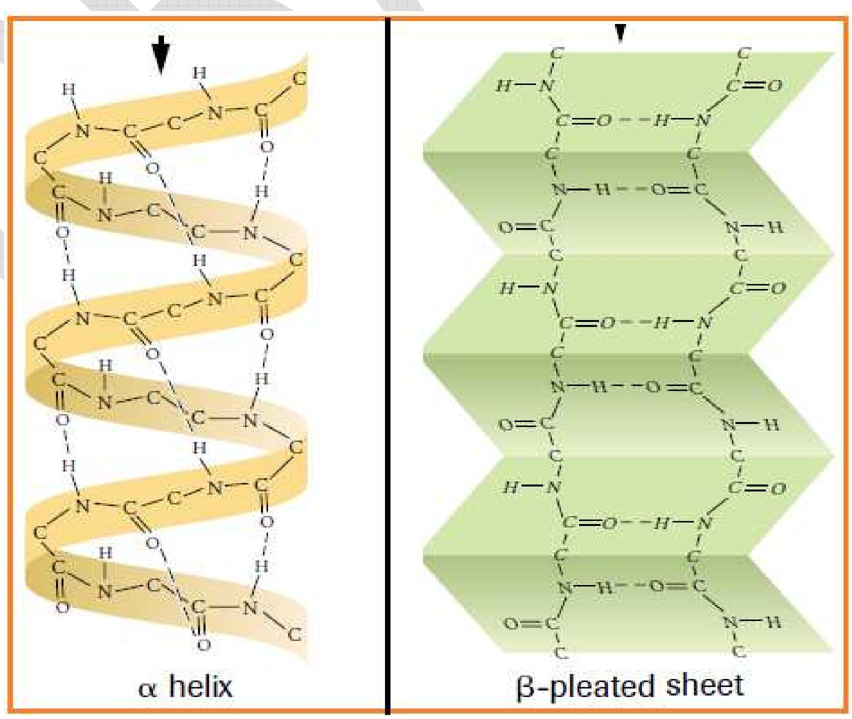
2) Secondary structure

All protein molecules are not simple long polymeric chains. Instead the peptide backbone of the protein structure will fold onto itself, to give proteins their unique shape. This folding of the polypeptide chains happens due to the interaction between the carboxyl groups along with the amine groups of the peptide chains, resulting in the formation of hydrogen bonds.

There are two kinds of shapes formed in the secondary structure. These are

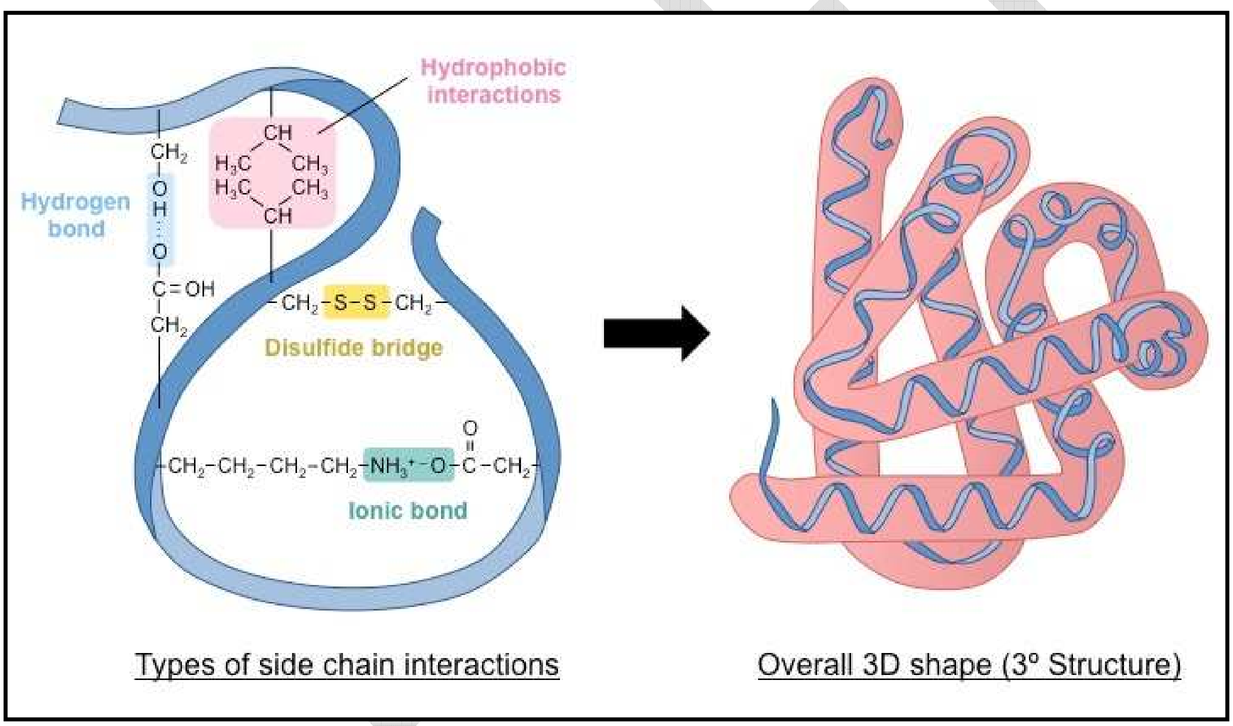
- **α-helix**: The backbone follows a helical structure. The hydrogen bonds with the oxygen between the different layers of the helix, giving it this helical structure.
- **β-pleated sheet**: here the polypeptide chains are stacked next to each other and their outer hydrogen molecules form intramolecular bonds to give it this sheet-like structure

The secondary structure of protein is a characteristic of fibrous and contractile proteins such as keratin of hair, fibrin of blood clots, myosin of muscle, collagen and elastin of connective tissue.



3) Tertiary structure

This is the structure that gives protein the 3-D shape and formation. After the amino acids form bonds (secondary structure) and shapes like helices and sheets, the structure can coil or fold at random. This is what we call the tertiary structure of proteins. It is a compact structure with hydrophobic side chains held interior while the hydrophilic groups are on the surface of the protein molecule. This type of arrangements ensures stability of the molecule. Tertiary structure is maintained chiefly by 4 kinds of bonds – Hydrogen bonds, Ionic bonds, hydrophobic bonds and Disulphide bonds. If this structure is disrupted or disturbed a protein is said to be denatured which means it is chemically affected and its structure is distorted.

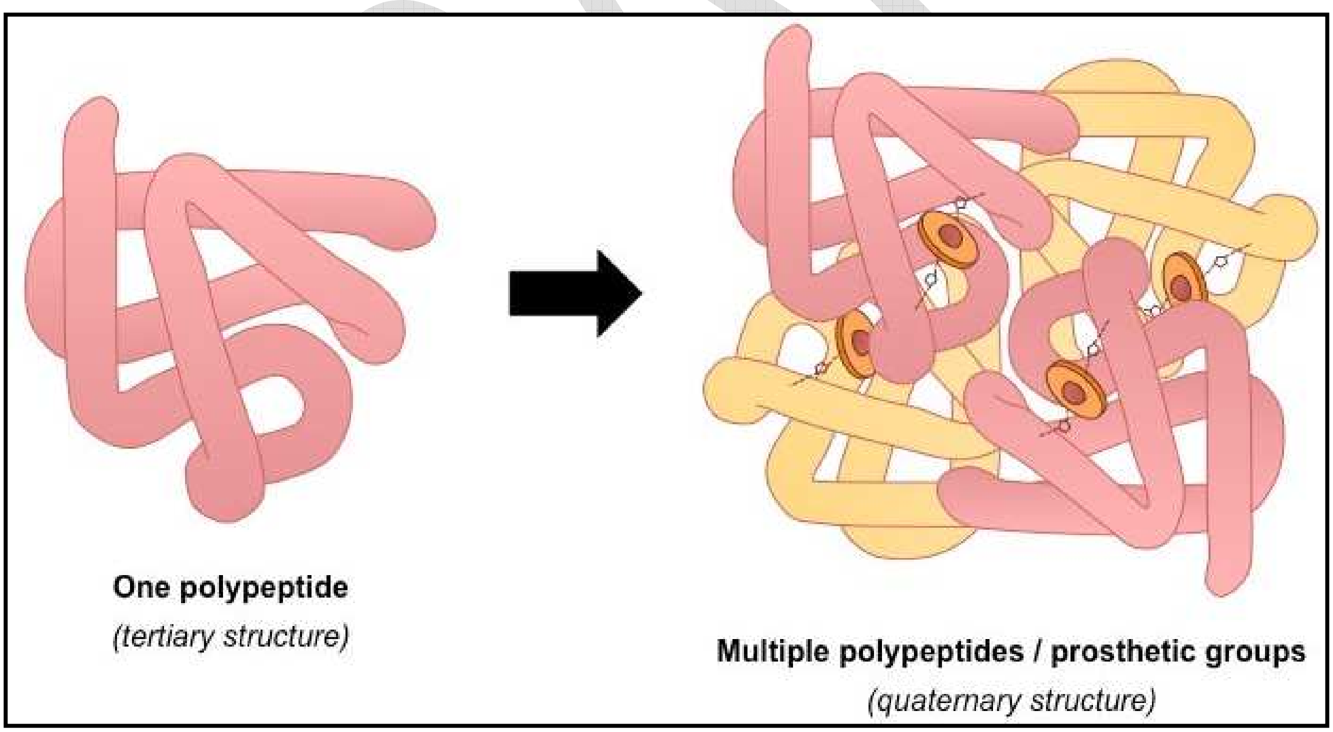


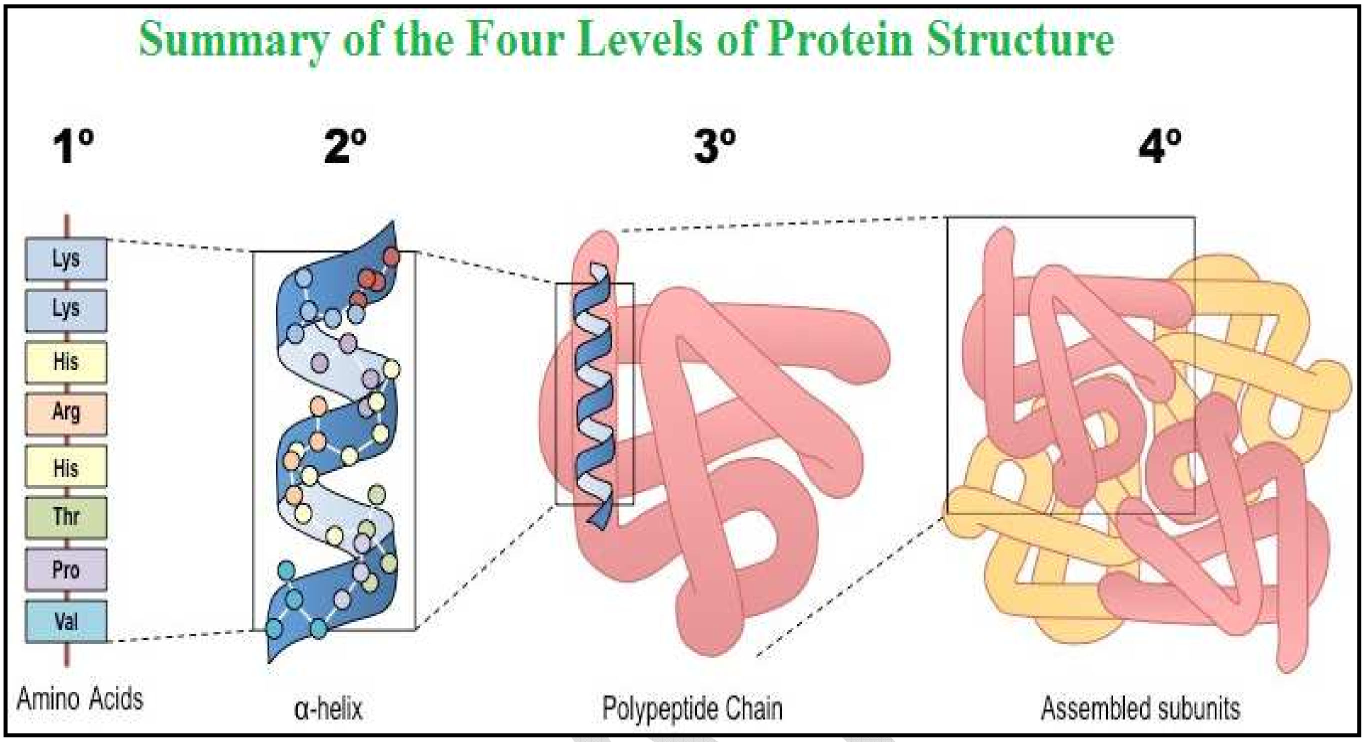
4) Quaternary structure

In proteins that are made up of two or more polypeptide chains, the quaternary structure refers to the specific orientation of these chains with respect to one another and the nature of the interactions that stabilize this orientation.

Many proteins contain more than one polypeptide chain. The individual polypeptides fold to yield secondary and tertiary structures but also bind to one another in precise ways through hydrogen bonds, vander waals forces, ionic bonds, disulphide bridges and hydrophobic interaction. The organization of two or more polypeptide chains into one unit is called the quaternary structure of protein.

It is important to note it is not necessary for proteins to have quaternary structures. Primary, secondary and tertiary structures are present in all natural proteins, but the same is not true for quaternary structure. Hence if a protein has only the first three structures it is considered to be a protein.




Summary of the Four Levels of Protein Structure
1°
2°
3°
4°
Lys
Lys
His
Arg
His
Thr
Pro
Val
Amino Acids
α-helix
Polypeptide Chain
Assembled subunits

OBJECTIVE TYPE QUESTIONS

1) The bond formed between the amino group of one amino acid and carboxylic group of other amino acid is called.
    a) Ionic bond
    b) Covalent bond
    c) Hydrogen bond
    d) **Peptide bond**
2) Which type of protein structure is shown in the myosin of muscle
    a) Primary
    b) **Secondary**
    c) Tertiary
    d) Quaternary
3) Which of the following statements about amino acids is not true
    a) They are the building blocks of proteins
    b) They contain amino and carboxylic groups
    c) They are used as nutritional supplements in foods
    d) Bond formed between the amino group of one amino acid and carboxylic group of other amino acid is covalent bond
4) Peptide bonds are present between
    a) Pyrimidine base
    b) **Amino acids**
    c) Purine bases
    d) Purine and Pyrimidine bases
5) Which of the following proteins was first sequenced by Sanger?

a) Myosin

b) Insulin

c) Myoglobin

d) Haemoglobin

6) How many amino acids make up a protein?

a) 10
b) 20
c) 30
d) 50

7) Which of the following cell organelles is involved in the process of protein synthesis?

a) Vesicles
b) **Ribosomes**
c) Synchrotrons
d) Mitochondria

8) Which of the following statements is true about the (primary) 1° structure of proteins?

a) The helical structure of the protein
b) Subunit structure of the protein
c) Three-dimensional structure of the protein
d) **The sequence of amino acids joined by a peptide bond**

9) Which of the following diseases is caused by protein deficiency?

a) Anaemia
b) **Kwashiorkor**
c) Hypothyroidism
d) All of the above

STRUCTURE OF DNA AND ITS DIFFERENCES WITH RNA

The genetic material in most organisms is DNA or Deoxyribonucleic acid; whereas in some viruses, it is RNA or Ribonucleic acid. A DNA molecule consists of two polynucleotide chains i.e. chains with multiple nucleotides. Let's understand the structure of this chain in detail.

Structure of Polynucleotide Chain

A nucleotide is made of the following components:

1) **Nitrogenous base** – These can be of two types (Purines and Pyrimidines)
   Purines include Adenine (A) and Guanine (G)
   Pyrimidines include Cytosine (C) and Thymine (T).
   In RNA, thymine is replaced by Uracil (U).

2) **Pentose sugar** – A pentose sugar is a 5-carbon sugar.
   In DNA, this sugar is deoxyribose
   In RNA, it is ribose

3) **Phosphate group**

Nitrogenous base + pentose sugar (via N-glycosidic linkage) = Nucleoside.

Nucleoside + phosphate group (via phosphoester linkage) = Nucleotide.

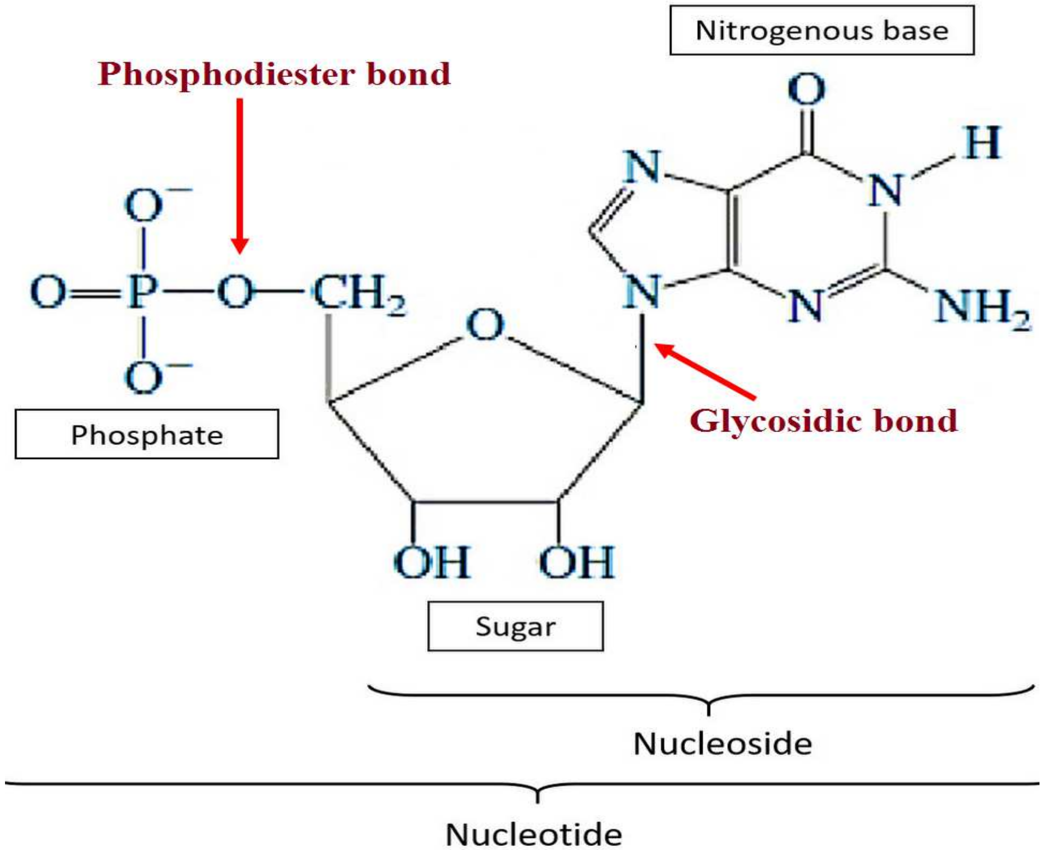
Nucleotide + Nucleotide (via 3′-5′ phosphodiester linkage) = Dinucleotide.

Many nucleotides linked together = Polynucleotide.

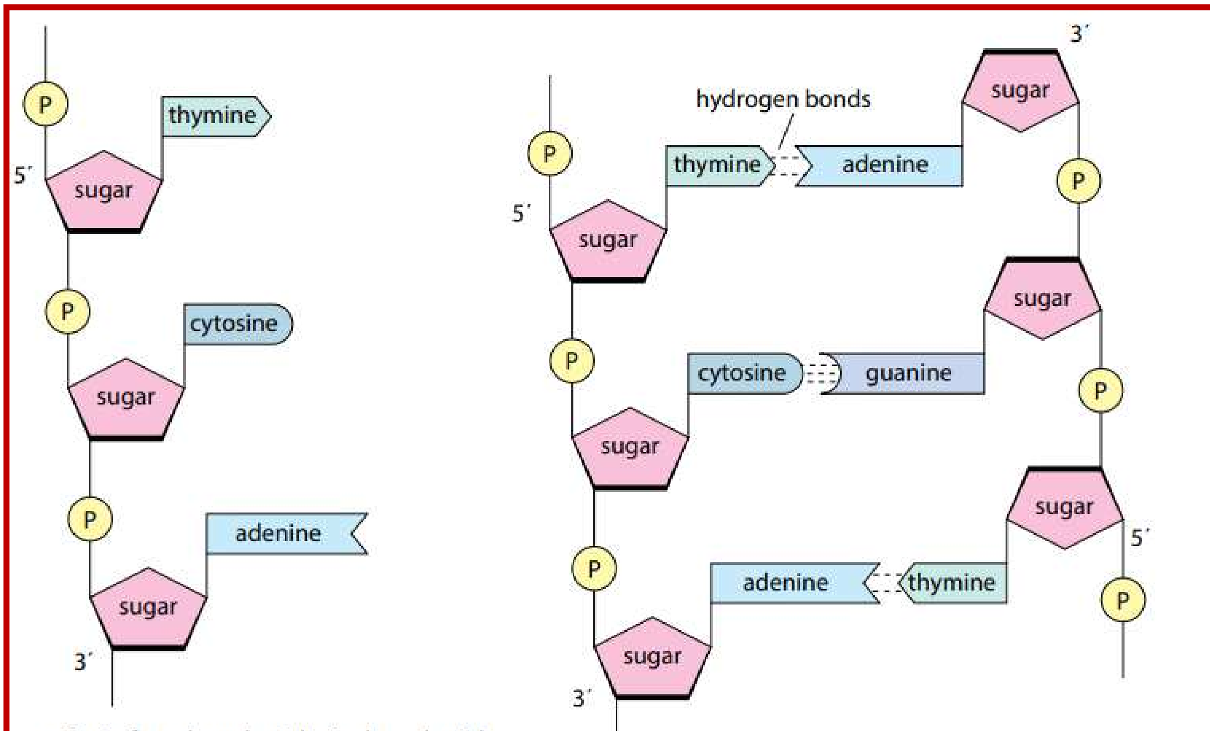
Chargaff's Rule

Erwin Chargaff, a Biochemist, found that the number of nitrogenous bases is present in equal amounts in the DNA. The amount of Adenine (A) is equal to the amount of Thymine (T). And the amount of Guanine (G) is equal to the amount of Cytosine (C).

The DNA of a cell contains equal amounts of purine and pyrimidine to keep the ratio 1:1.



A polynucleotide has a free phosphate group at the 5′ end of the sugar and this is called the 5′ end. Similarly, the sugar also has a free 3′-OH group at the other end of the polynucleotide which is called the 3′ end. The backbone of a polynucleotide chain consists of pentose sugars and phosphate groups; whereas the nitrogenous bases project out of this backbone.



a Part of a polynucleotide. Each nucleotide is linked to the next by covalent bonds between the phosphates and sugars.

b Part of a DNA molecule. Two polynucleotides, running in opposite directions, are held together by hydrogen bonds between the bases. A links with T by two hydrogen bonds; C links with G by three hydrogen bonds. This is complementary base pairing.

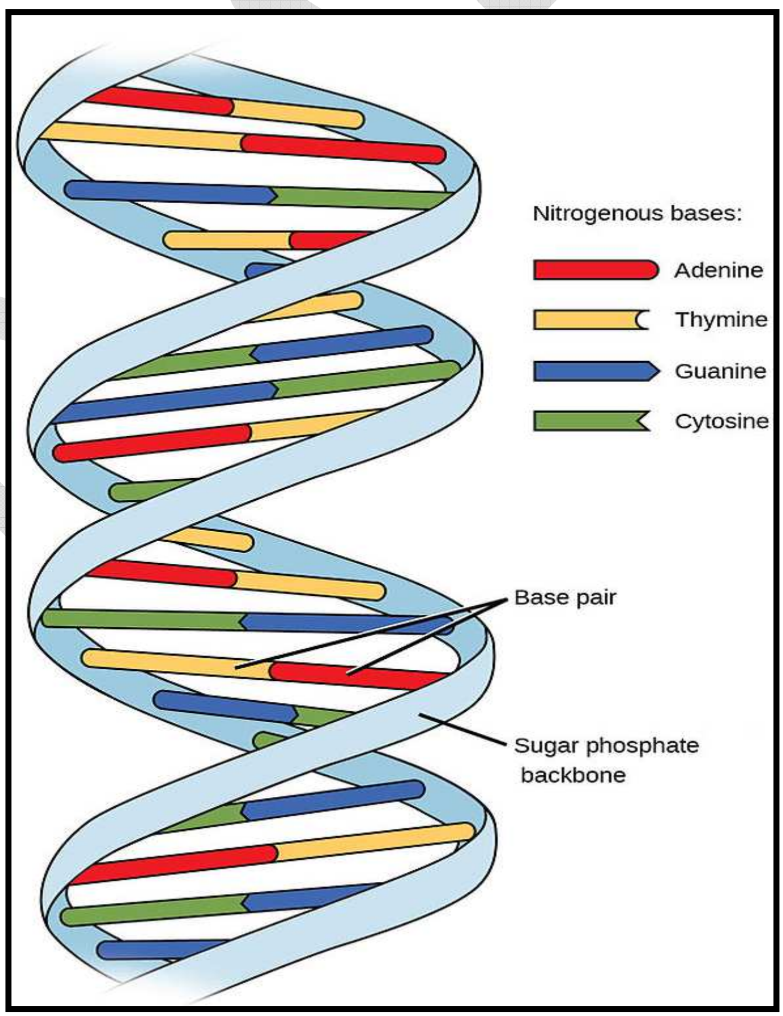
Double Helix Structure

DNA is a long polymer and therefore, difficult to isolate from cells in an intact form. This is why it is difficult to study its structure. However, in 1953, James Watson and Francis revealed the 'double helix' model of the structure of DNA, based on X-ray diffraction data from Maurice Wilkins and Rosalind Franklin.

This model also reveals a unique property of polynucleotide chains – Base pairing. It refers to the hydrogen bonds that connect the nitrogen bases on opposite DNA strands. This pairing gives rise to complementary strands i.e. if you know the sequence of bases on one strand, you can predict the bases on the other strand. Additionally, if each DNA strand acts as a template for synthesis (parent) of a new strand, then the new double-stranded DNA (daughters) produced are identical to the parental DNA strand.

Salient Features of DNA Double-Helix

- It consists of two polynucleotide chains where the sugar and phosphate group form the backbone and the nitrogenous bases project inside the helix.
- The two polynucleotide chains have anti-parallel polarity i.e. if one strand has $5' \rightarrow 3'$ polarity, the other strand has $3' \rightarrow 5'$ polarity.
- The bases on the opposite strands are connected through hydrogen bonds forming base pairs (bp). Adenine always forms two hydrogen bonds with thymine from the opposite strand and vice-versa. Guanine forms three hydrogen bonds with cytosine from the opposite strand and vice-versa. Therefore, a purine always pairs with a pyrimidine on the other strand, giving rise to a uniform distance between the two strands of the helix.
- The two strands coil in a right-handed fashion. Each turn of the helix is 3.4nm (or 34 Angstrom units) consisting of 10 nucleotides. These nucleotides are at a distance of 0.34nm (or 3.4 Angstrom units).
- The helix is stable because of the base pairs that stack over one another and hydrogen bonds that hold the bases together.



There are three different DNA types:

1. **A-DNA**: It is a right-handed double helix similar to the B-DNA form.
2. **B-DNA**: This is the most common DNA conformation and is a right-handed helix.
3. **Z-DNA** is a left-handed DNA where the double helix winds to the left in a zig-zag pattern.

FUNCTIONS of DNA

1. **Replication:** DNA has the ability to replicate. (formation of new molecules of daughter DNA from parent DNA).The replication of DNA takes place at every cell division to ensure the equal distribution of DNA in daughter cells. Through its replication, DNA acts as the key to heredity.

2. **Synthesis of RNA (Transcription):** All the three types of RNA (m RNA, r RNA and t RNA) which take part in protein synthesis, are synthesized under the control of DNA by the process known as transcription.

3. **Synthesis of proteins (Translation):** Translation is the guiding of amino acid sequence in proteins. Thus, DNA controls protein synthesis via RNA. Due to this ability of DNA, specific proteins and enzymes are synthesized by the cell.

   $$\text{DNA} \xrightarrow{\text{Transcription}} \text{RNA} \xrightarrow{\text{Translation}} \text{Proteins}$$

   This scheme is called the "central Dogma" of molecular biology. By so controlling protein manufacture, DNA ultimately controls the entire structural and functional make up of every cell.

4. **DNA and mutations:**

   Under certain conditions, the nitrogen base sequence of a particular amino acid gets altered. Such alterations then are stable and persist into succeeding molecular generations of DNA. When such changes occur, the structural and functional traits of a cell also change correspondingly, leading to mutations. Through changes in its cells, a whole animal and its progeny may thus become changed in the cause of successive generations; this is equivalent to evolution.

RNA – differences with DNA

Parameter	DNA	RNA
Location	Located in the nucleus of a cell and in the mitochondria.	Located in the cytoplasm, nucleus, and in the ribosome.
Structure	DNA is a double-stranded molecule consisting of a long chain of nucleotides. B type of helix.	It is a single-stranded helix consisting of a short chain of nucleotides. A type of helix.
Composition	Deoxyribose sugar phosphate backbone adenine, guanine, cytosine, thymine bases.	Ribose sugar phosphate backbone adenine, guanine, cytosine, uracil bases.
Bases Pairing	GC (Guanine pairs with Cytosine) A-T (Adenine pairs with Thymine).	GC(Guanine pairs with Cytosine) A-U(Adenine pairs with Uracil)
Function	Transmits genetic information to make other cells and new organisms. Long-term storage of genetic information	It transfers the genetic code from the nucleus to the ribosomes to make proteins.
Propagation	DNA is self-replicating.	Synthesized from DNA.
Stability	DNA is a more stable molecule than RNA. DNA is stable under alkaline conditions.	Much more reactive than DNA and is not stable in alkaline conditions.

Multiple choice questions

1. DNA strands are antiparallel because of

   A. H bonds

   B. Phospho-diester bonds

   C. disulphide bonds

   D. Phospho-ester bonds

2. The number of base pairs in one complete turn of DNA helix is

   A. 12

   B. 10

   C. 11

   D. 8

3. What is a "double helix"?

   A. Two Y-shaped strand

   B. Two loops, like a figure eight

   C. Two spirals, like a twisty ladder

   D. Two X-shaped strands

4. DNA is constructed by repeating units. What are these units called?

   A. Amino acids

   B. Nucleotides

   C. Monosaccharides

   D. Fatty acids

5. To whom is the discovery of the structure of DNA credited?

   A. Gregor Mendel

   B. Charles Darwin

C. James Watson

D. Hargobind Khorana

6. What part of DNA is found at the centre of the molecule?

A. Amino acids

B. Nitrogen Bases

C. Phosphates

D. Sugars

7. What process is used to copy DNA to produce another identical strand?

A. Replication

B. Translation

C. Transcription

D. Annealing

8. The most important molecule involved in long-term storage of genetic information is

A. protein

B. lipid

C. deoxyribonucleic acid or DNA

D. ribonucleic acid or RNA

9. The primary chemical structure of DNA includes

A. a phosphorylated sugar backbone

B. Individual components called amino acids

C. Individual components called triglycerides

D. Complete structural symmetry

10. Double-stranded DNA, when it occurs, is

A. Antiparallel

B. parallel

C. complementary

D. Both A and C

11. The base pairing arrangements that occur in double-stranded DNA as determined by hydrogen bonding include

A. A with C and G with T

B. A with U and G with C

C. A with T and G with C

D. C with T and A with G

12. The complementary strand of DNA strand having nucleotide sequence as GAT CAA is

A. TTGATG

B. AGAAUU

C. CUAGUU

D. CTAGTT

13. The diameter of DNA molecule is

A. 20 A^0

B. 50 A^0

C. 100 A^0

D. 200 A^0

RNA bits